



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-13: 1 Class



*Please listen Carefully
&
Keep your phone(s) on silent mode*

Concept 11.3: Transduction: Cascades of molecular interactions relay signals from receptors to target molecules in the cell

- Signal transduction usually involves multiple steps
- Multistep pathways can **amplify a signal**: A few molecules can produce a large cellular response
- Multistep pathways provide **more opportunities for coordination and regulation** of the cellular response

Signal Transduction Pathways

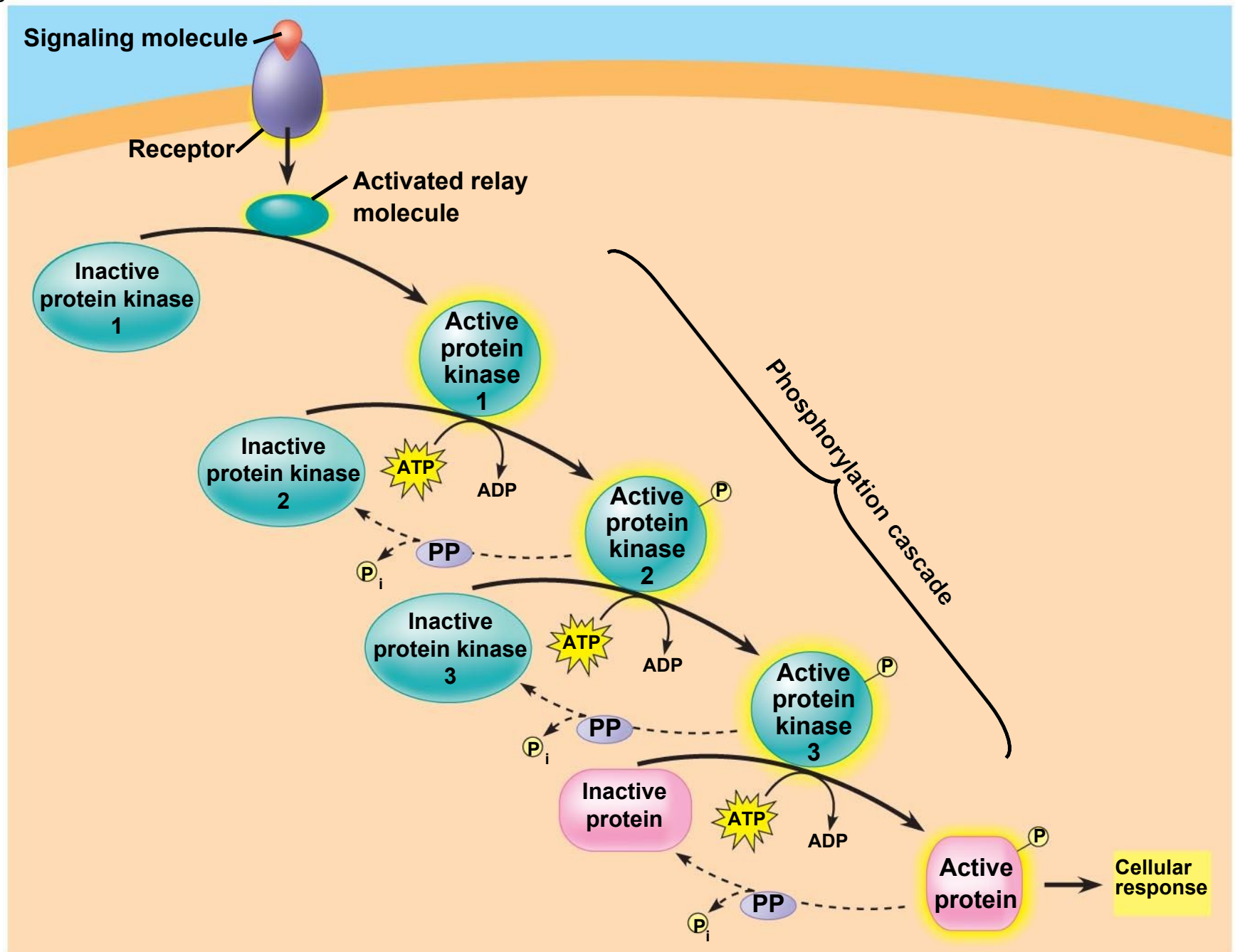
- The molecules that relay a signal from receptor to response are mostly **proteins**
- Like falling dominoes, the receptor activates another protein, which activates another, and so on, until the protein producing the response is activated
- At each step, the signal is transduced into a different form, usually a shape change in a protein

Protein Phosphorylation and Dephosphorylation

- In many pathways, the signal is transmitted by a cascade of protein phosphorylations
- **Protein kinases** transfer phosphates from ATP to protein, a process called phosphorylation

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- **Protein phosphatases** remove the phosphates from proteins, a process called **dephosphorylation**
 - This phosphorylation and dephosphorylation system acts as a molecular switch, turning activities on and off

Fig. 11-9



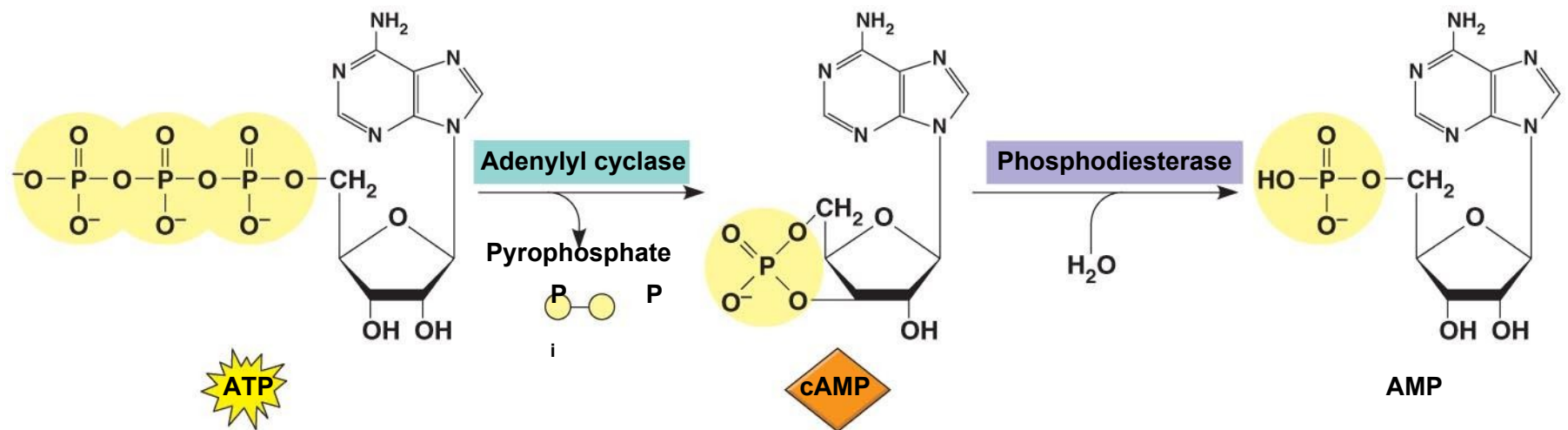
Small Molecules and Ions as Second Messengers

- The extracellular signal molecule that binds to the receptor is a pathway's “first messenger”
- **Second messengers are small, nonprotein, water-soluble molecules or ions that spread throughout a cell by diffusion**
- Second messengers participate in pathways initiated by G protein-coupled receptors and receptor tyrosine kinases
- **Cyclic AMP and calcium ions are common second messengers**

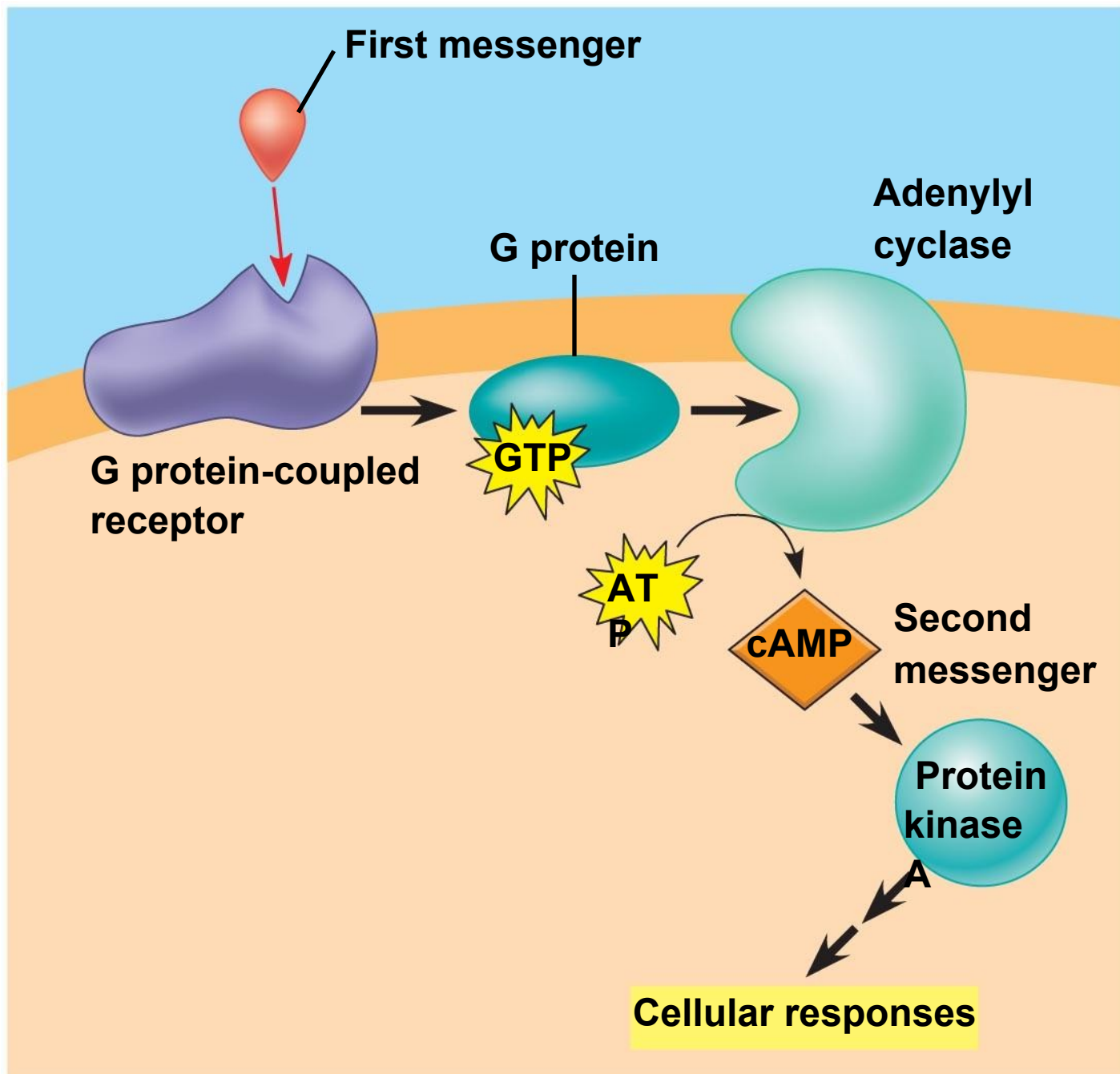
Cyclic AMP

- **Cyclic AMP (cAMP)** is one of the most widely used second messengers
- **Adenylyl cyclase**, an enzyme in the plasma membrane, **converts ATP to cAMP in response to an extracellular signal**

Fig. 11-10



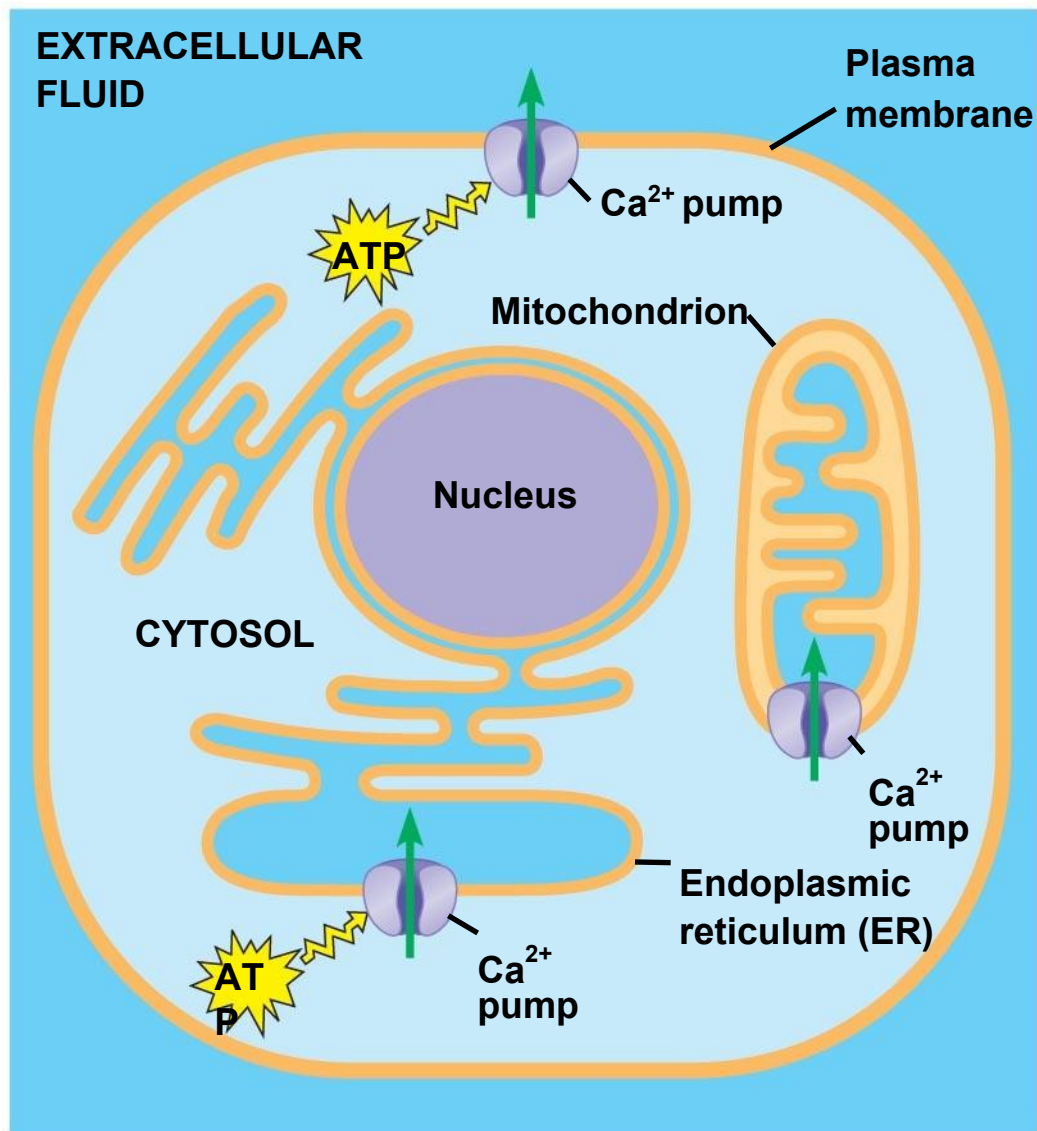
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- Many signal molecules trigger formation of cAMP
 - Other components of cAMP pathways are G proteins, G protein-coupled receptors, and protein kinases
 - cAMP usually activates protein kinase A, which phosphorylates various other proteins
 - Further regulation of cell metabolism is provided by G-protein systems that inhibit adenylyl cyclase



Calcium Ions and Inositol Triphosphate (IP₃)

- Calcium ions (Ca^{2+}) act as a second messenger in many pathways
- Calcium is an important second messenger because cells can regulate its concentration

Fig. 11-12



Key

- High [Ca²⁺]
- Low [Ca²⁺]

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- A signal relayed by a signal transduction pathway may trigger an increase in calcium in the cytosol
 - Pathways leading to the release of calcium involve **inositol triphosphate (IP₃)** and **diacylglycerol (DAG)** as additional second messengers

Fig. 11-13-1

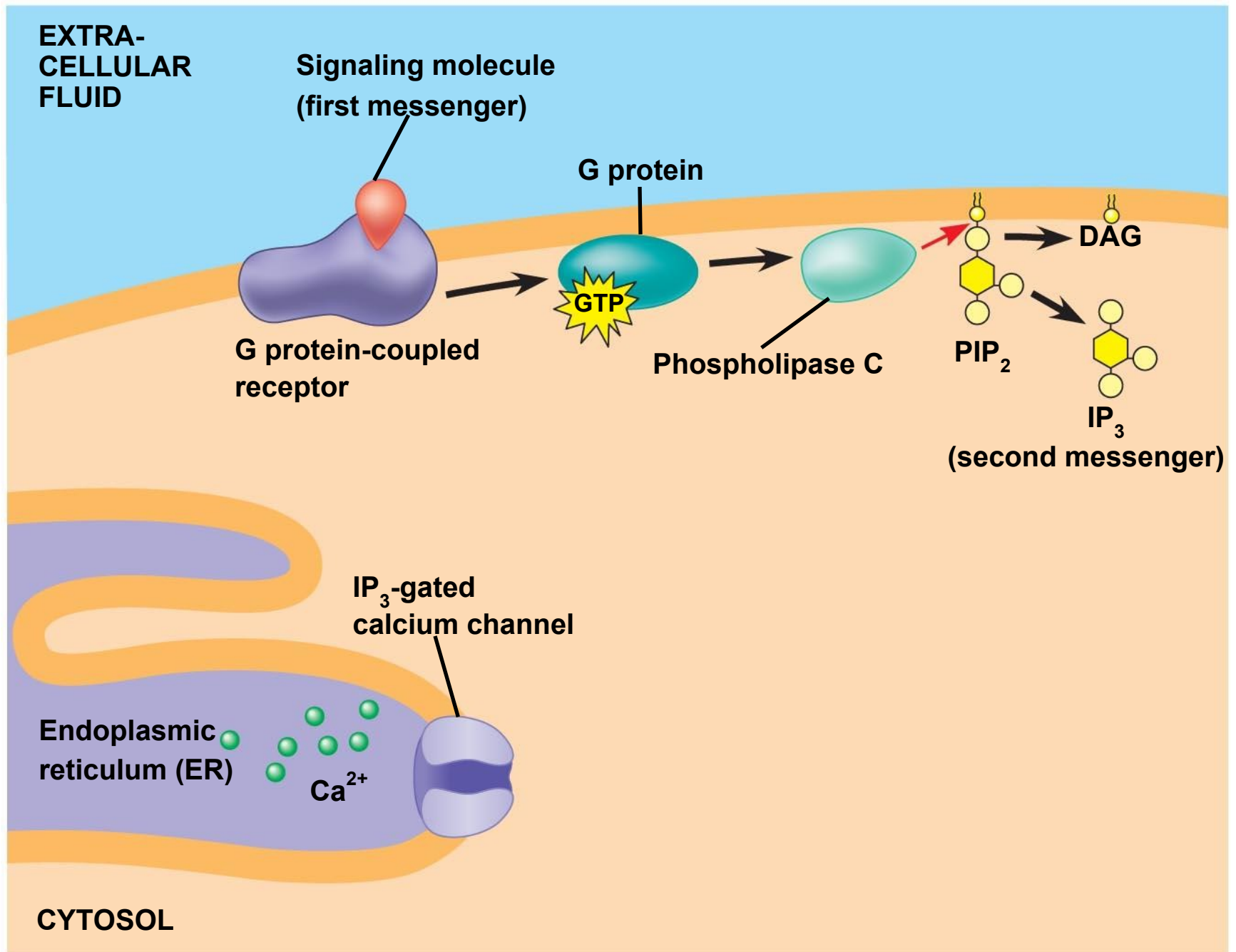


Fig. 11-13-2

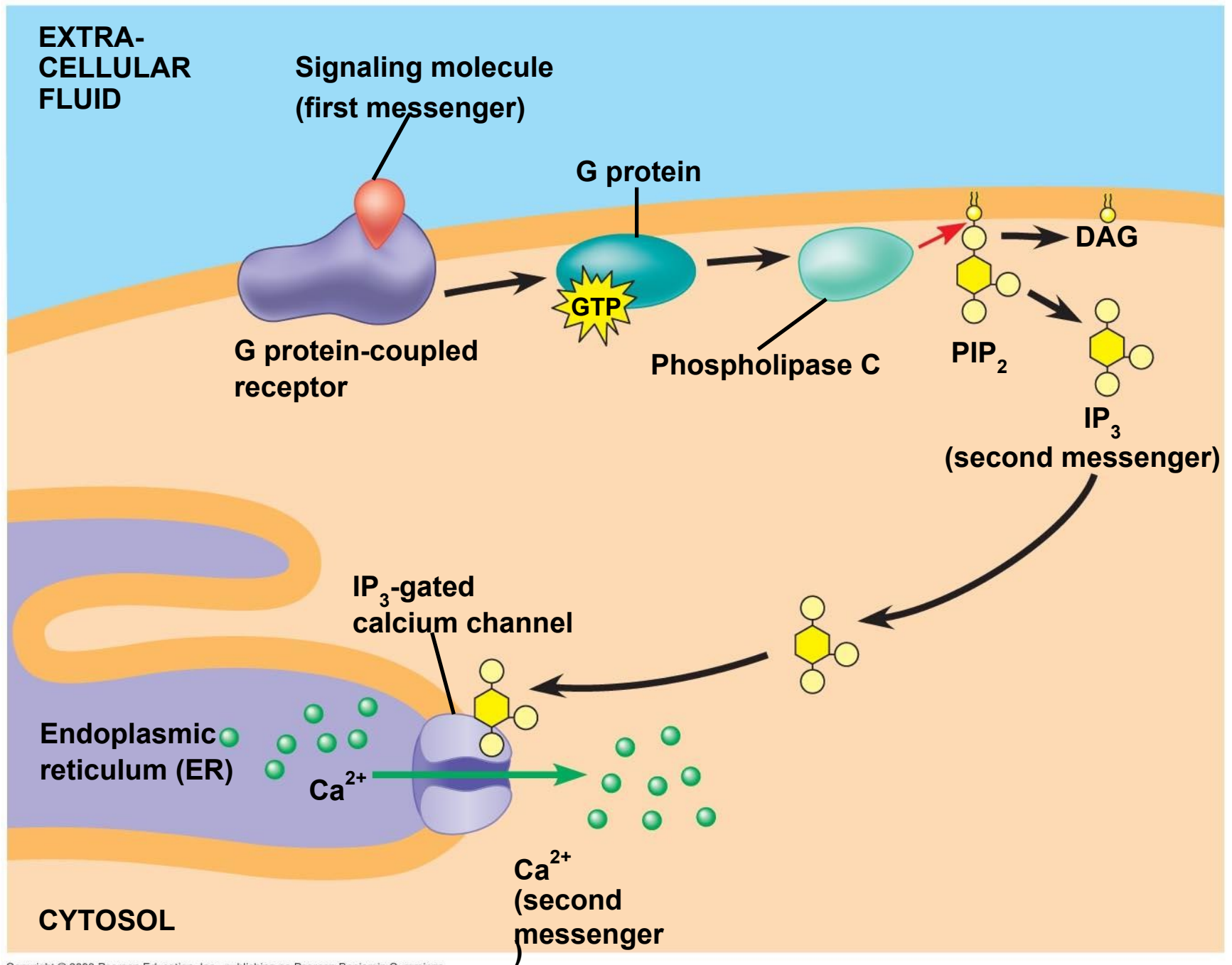
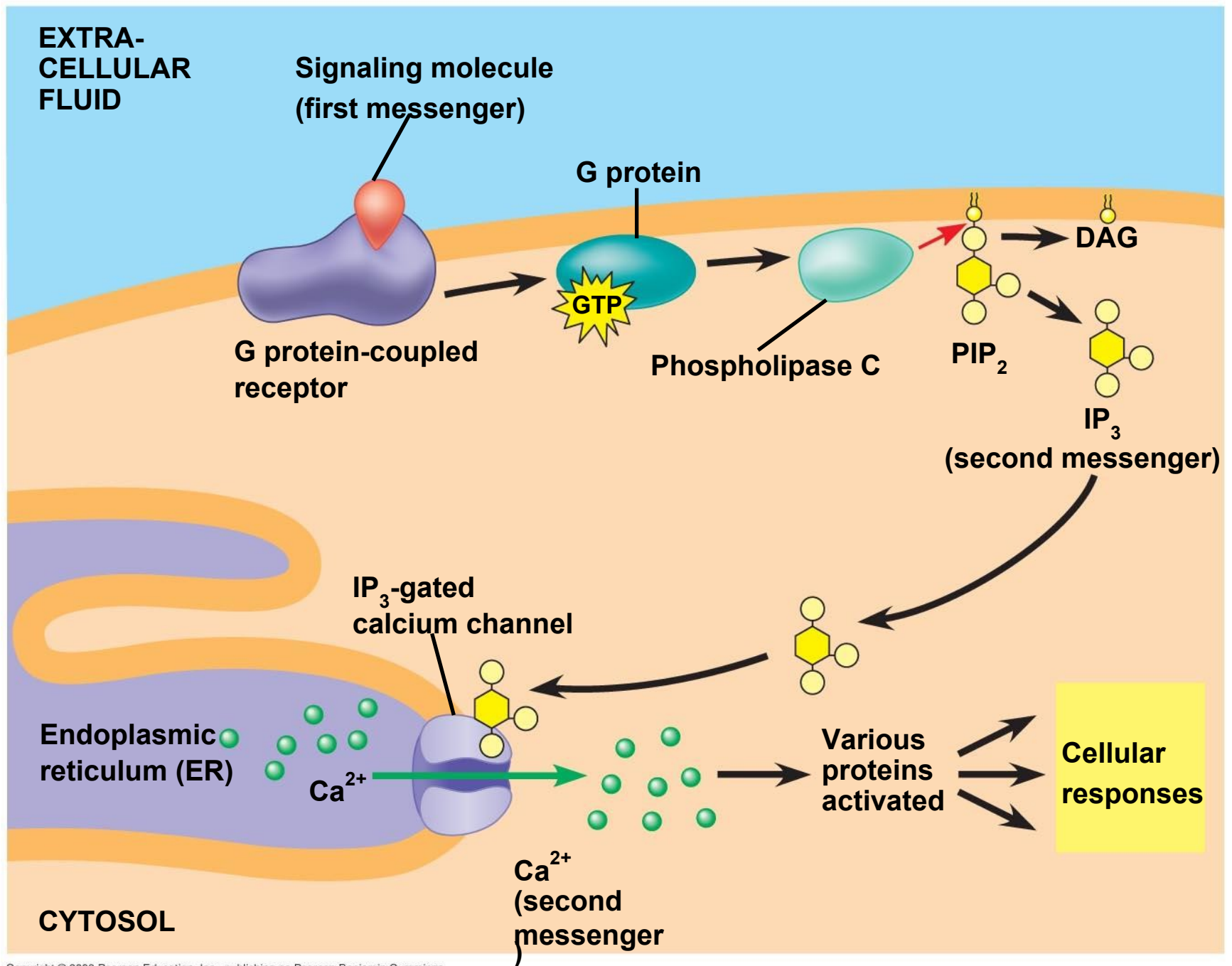


Fig. 11-13-3



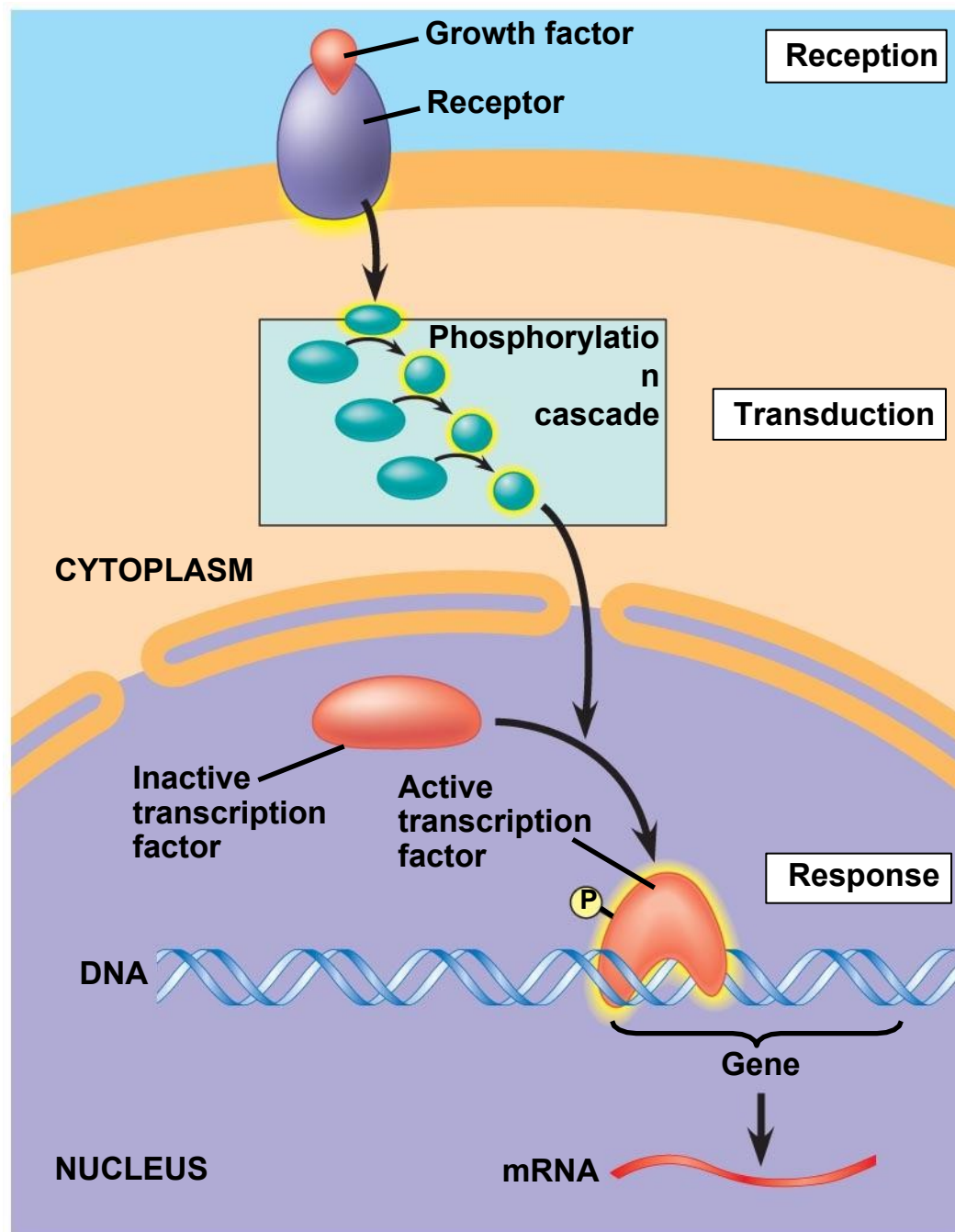
Concept 11.4: Response: Cell signaling leads to regulation of transcription or cytoplasmic activities

- The cell's response to an extracellular signal is sometimes called the “output response”

Nuclear and Cytoplasmic Responses

- Ultimately, a signal transduction pathway leads to regulation of one or more cellular activities
- The response may occur in the cytoplasm or may involve action in the nucleus
- Many signaling pathways regulate the *synthesis* of enzymes or other proteins, usually by turning genes on or off in the nucleus
- The final activated molecule may function as a transcription factor

Fig. 11-14



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- Other pathways regulate the *activity* of enzymes

Fig. 11-15

Reception

Binding of epinephrine to G protein-coupled receptor (1 molecule)



Transduction

Inactive G protein

Active G protein (10^2 molecules)

Inactive adenylyl cyclase

Active adenylyl cyclase (10^2)

ATP

Cyclic AMP (10^4)

Inactive protein kinase A

Active protein kinase A (10^4)

Inactive phosphorylase kinase

Active phosphorylase kinase (10^5)

Inactive glycogen phosphorylase

Active glycogen phosphorylase (10^6)

Response

Glycogen
Glucose-1-phosphate
(10^8 molecules)